Size control: vertex-cap baseline against BudgetMesh

21 scenes, one target per dataset: 10 MB (13 captured scenes), 2.5 MB (8 objects). Same codec, same fine-tuning, one L40S GPU. All numbers are measured runs. 13 Nov 2026.

1 Result

	Vertex cap	BudgetMesh
Scenes within $\pm 10\%$ after 1 training	15 / 21	20 / 21
Scenes within $\pm 10\%$ after 2 trainings	17 / 21	–
Scenes within $\pm 10\%$ after 3 trainings	18 / 21	–
Scenes never within $\pm 10\%$	3	1
Scenes within $\pm 2\%$ after 1 training	3 / 21	19 / 21
Size error, range over scenes (1 training)	$-39.6 \ldots +16.3\%$	$-22.6 \ldots +0.7\%$
Slowest scene that reached $\pm 10\%$: GPU minutes	81	61
Slowest scene that reached $\pm 10\%$: trainings	2	1

Time is training + fine-tuning + encoding. Slowest scenes: bonsai (cap), counter (BudgetMesh). BudgetMesh never retrains.

2 Every scene that needed a retry, training by training

scene (target)	training	cap	vertices kept	file (MB)	error	BudgetMesh
never within $\pm 10\%$						
drjohnson (10)	1	1,081,000	881,894	11.57	$+15.7\%$	10.05 MB, $+0.5\%$
	2	902,000	900,413	11.69	$+16.9\%$	
mic (2.5)	1	174,000	106,744	2.04	-18.4%	2.49 MB, -0.4%
	2	247,000	107,011	2.05	-18.0%	
ficus (2.5)	1	171,000	68,358	1.51	-39.6%	1.93 MB, -22.6%
	2	515,000	68,896	1.50	-40.0%	
reached $\pm 10\%$ *after a retry*						
kitchen (10)	1	1,086,000	956,939	11.63	$+16.3\%$	10.07 MB, $+0.7\%$
	2	900,000	716,750	9.02	-9.8%	
bonsai (10)	1	1,073,000	754,307	8.32	-16.9%	9.99 MB, -0.1%
	2	1,367,000	934,609	9.86	-1.4%	
lego (2.5)	1	183,000	156,040	2.79	$+11.6\%$	2.49 MB, -0.4%
	2	154,000	156,438	2.79	$+11.7\%$	
	3	130,000	133,085	2.50	$+0.2\%$	

drjohnson: a 17% smaller cap kept more vertices. mic, ficus: the cap does not bind; raising it changes nothing. lego: a 16% smaller cap gave the same mesh. Ficus cannot fill 2.5 MB with any method.

3 How the baseline is set

Method. MeshSplatting trained with its own `-max_points` option (the vertex cap), then the same fine-tuning and the same codec as BudgetMesh.

Prior (how the cap is chosen). File size is fitted as a line in the cap, MB $= a \cdot$ cap $+ c$, by least squares on cap runs of *other* scenes of the same kind; the test scene is left out. Then cap $= (T - c)/a$, rounded to 1,000.

- Captured scene: the 60 cap runs of the other 10 Mip-NeRF 360 / T&T scenes (6 caps each, 0.2–3.1 M). Deep Blending: all 66 runs of the 11 scenes. $a = 7.2$–7.4 MB per million, $c = 2.1$–2.2 MB. Caps chosen: 1,069,000–1,105,000.
- Object: the 49 cap runs of the other 7 objects. $a = 8.5$–9.1 MB per million, $c = 0.9$–1.0 MB. Caps chosen: 171,000–183,000.

Retry (fixed before the runs). If the file misses the target by more than 10%: $\text{cap}_2 = \text{cap}_1\,(T - c)/(\text{MB}_1 - c)$, train again from scratch.

	baseline	scenes	result (size error)
A	cap, same-kind prior, 1 training	21	15 within $\pm 10\%$; $-39.6 \ldots +16.3\%$
B	A + retry on a miss	6 retried	17 within $\pm 10\%$
C	B + third training	lego	18 within $\pm 10\%$
D	cap, prior from the other kind (captured $\rightarrow$ lego)	1	-41.8%; retrained: -21.4%
E	cap rescaled by target / measured size, retrained after a hit	3	garden $+3.0 \rightarrow +16.4\%$, truck $-0.5 \rightarrow 0.0\%$, playroom $+9.6 \rightarrow +16.3\%$
F	our size measurement and growth plan, MeshSplatting's triangle choice, no collapse	4	-1.7, -7.2, $+12.0$, $+9.6\%$
G	BudgetMesh, previous controller	21	18 within $\pm 10\%$; $-21.6 \ldots +13.5\%$
H	BudgetMesh, final design	21	20 within $\pm 10\%$; $-22.6 \ldots +0.7\%$

4 Sensitivity of the cap

166 cap runs in total (96 on captured scenes, 70 on objects). Same code, options and random seed in every run; only the cap differs.

pairs of caps on the same scene, differing by at most	1%	5%	25%
pairs	11	29	123
larger cap gave a smaller or equal file	4	12	35
file changed by more than 10%	2	8	42
file change, range	$-23.7 \ldots +6.8\%$	$-23.7 \ldots +14.6\%$	$-23.7 \ldots +41.5\%$

garden, cap	1,073,000	1,074,000	1,104,000	1,105,000	1,106,000
file (MB)	11.64	11.61	13.30	10.30	10.15
vertices kept / cap	0.94	0.93	1.06	0.80	0.79
playroom, cap	986,000	1,080,000	1,081,000	1,082,000	1,201,000
file (MB)	11.64	10.37	10.96	11.07	14.60
vertices kept / cap	0.95	0.77	0.78	0.78	0.96
BudgetMesh on garden, target 10 MB, seeds 0 / 1 / 2			10.21	10.16	10.16 MB

- **Size of the effect.** garden: cap $1{,}104{,}000 \rightarrow 1{,}105{,}000$ $(+0.09\%)$ moves the file $13.30 \rightarrow 10.30$ MB (-23%).
- **Direction.** In 12 of 29 pairs of caps within 5% of each other, the larger cap gave the smaller (or the same) file.
- **Range.** Vertices kept after training: 0.68–1.06 of the cap on captured scenes, 0.13–1.05 on objects.
- **Why.** The cap bounds growth; stitching, opacity pruning and final cleaning act afterwards and remove a share the cap does not control. When growth stops below the cap (mic, ficus), the cap has no effect at all.

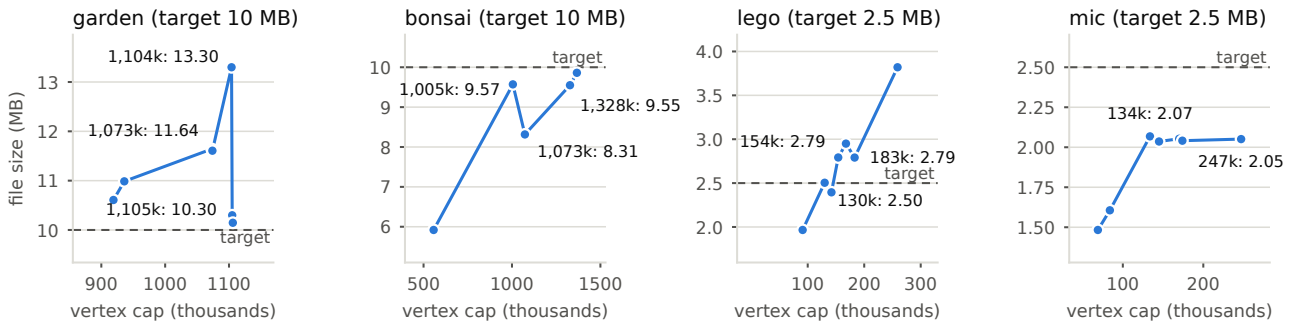


Figure 1: File size against the vertex cap, all cap runs of four scenes in the shown range. Labels: cap, file size in MB.

Limits of this report. One target per dataset. The dense caps on garden and playroom were placed next to caps that had already behaved oddly, so the pair counts show that such jumps exist, not how often a random cap hits one. Cap runs use one seed; six cap trainings from earlier batches have no stage time stamps and are timed with the class mean (34 min captured, 18 min objects).